Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.				award (1) each for up to 9 of following white precipitate indicates the unknown halogen is chlorine (1) n(C) = 3.25 n(H) = 3.22 n(Cl) = 1.63 (1) for all three ratio of C:H:Cl is 2:2:1 therefore empirical formula is C_2H_2Cl (1) molecular ion peak at 122 therefore molecular formula is $C_4H_4Cl_2$ (1) n(Br_2) = 2 mol (1) n(**A**) = 1 mol (1) bromine reacts with **A** suggests a C=C double bond (1) 1 mol of **A** reacts with 2 mol of Br_2 ⇨ two double bonds (1) 1 peak in ^{1}H NMR spectrum ⇨ one hydrogen environment (1) 2 peaks in ^{13}C NMR spectrum ⇨ two carbon environments (1) IR peak at approx. 650 cm^{-1} is C—Cl IR peak at approx. 1650 cm^{-1} is C=C (1) for either mass spec peak at 87 is $C_4H_4Cl^+$ mass spec peak at 52 is $C_4H_4^+$ peaks in ratio of 9:6:1 around 122 indicates two Cl atoms (1) for any of three award (1) for identification of correct compound 2,3-dichlorobuta-1,3-diene (accept correct structure)	3	3	4	10	3	2
				Question 8 total	**3**	**3**	**4**	**10**	**3**	**2**